# Irene Le

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## EDUCATION

**B.S. Computer Science** | University of California, Santa Cruz — *Expected May 2027*

**A.S. CyberDefense** | Saddleback College — *Expected May 2027*

## CERTIFICATIONS

**CompTIA A+**

## TECHNICAL SKILLS

**Languages:** C, C++, Python (PyTorch, Pandas, NumPy, scikit-learn), SQL, JavaScript, HTML/CSS
**Tools & Platforms:** Git, Linux/Unix, Bash/Shell Scripting, Wireshark, GDB, Docker, Visual Studio Code
**Concepts:** Data Structures & Algorithms, Multi-threading, POSIX Threads, TCP/IP, Socket Programming, REST APIs, OOP, Machine Learning, Debugging, Software Integration & Testing, Agile

## WORK EXPERIENCE

**Programming Instructor** | The Coder School | Cerritos, CA — *Feb 2025 – August 2026*

- Develop and deliver project-based curricula in Python and C++, emphasizing algorithmic thinking, clean code practices, and real-world debugging techniques
- Increased student engagement by 30% through hands-on labs and live coding sessions; created documentation and lesson plans tailored to varied skill levels
- Conduct 1:1 debugging sessions resolving logic, syntax, and runtime errors, functioning as a technical mentor and code reviewer for 50+ students

**Software Mentor & Lab Assistant** | Baskin School of Engineering UCSC | Santa Cruz, CA — *Aug 2023 – Jan 2024*

- Mentored 100+ students in C/C++ data structures and algorithms; diagnosed and resolved hundreds of code defects, driving a 30% increase in assignment completion rates
- Facilitated 10+ group office hours sessions per quarter through collaborative problem-solving and live debugging
- Reviewed and graded 400+ code submissions for correctness, efficiency, and style — contributing to a 25% improvement in overall code quality

**Computer Analyst** | Niem Tin Viet, LLC | Garden Grove, CA — *Aug 2020 – June 2022*

- Provided hands-on technical support for office staff including PC setup, configuration, and troubleshooting of hardware and software issues
- Tracked and maintained client financial account data in QuickBooks and Excel, ensuring accuracy and organization across multiple records
- Diagnosed and resolved day-to-day technology issues to minimize downtime and keep office operations running smoothly

## PROJECTS

**Encrypted File Sharing CLI** | *C, POSIX Sockets, OpenSSL, Linux*

- Built a peer-to-peer file transfer tool over raw TCP sockets using OpenSSL (TLS + AES-256); multi-threaded server handles concurrent sessions with SHA-256 integrity validation
- Documented threat model covering man-in-the-middle and replay attack mitigations, demonstrating security-first engineering mindset

**Multi-Threaded HTTP Server** | *C, POSIX Threads*

- Designed and built an HTTP server handling concurrent requests via POSIX system calls; thread-safe circular queue supports configurable worker pool and atomic audit logging
- Implemented linearized audit log ensuring correctness and traceability across all concurrent connections

**Network Packet Analysis Tool** | *C, POSIX Threads, Socket Programming*

- Built a multi-threaded packet analyzer using socket programming to capture live traffic; producer-consumer architecture with thread-safe logging and TCP/IP header parsing
- Generated per-session throughput and protocol breakdown reports for network diagnostics and security monitoring

**Library Management System** | *C++*

- Developed a console-based C++ application with persistent file-backed storage for books and members; utilized arrays, vectors, and linked lists for in-memory data management
- Built a menu-driven interface with robust error handling, input validation, and modular design supporting full CRUD operations